1. Install and Setup java environment

Install the below files as follows
1) jre-8u421-windows-x64
2) jdk-22_windows-x64_bin
3) Eclipse

All three software's are available in this link : google drive / Download and install Java / https://drive.google.com/drive/u/3/folders/1BED8nQ7ymh5vhRmrbiqb2ANbQZ2UJCMb

Install Java on Windows 10
After downloading the installation file, proceed with installing Java on your Windows system.

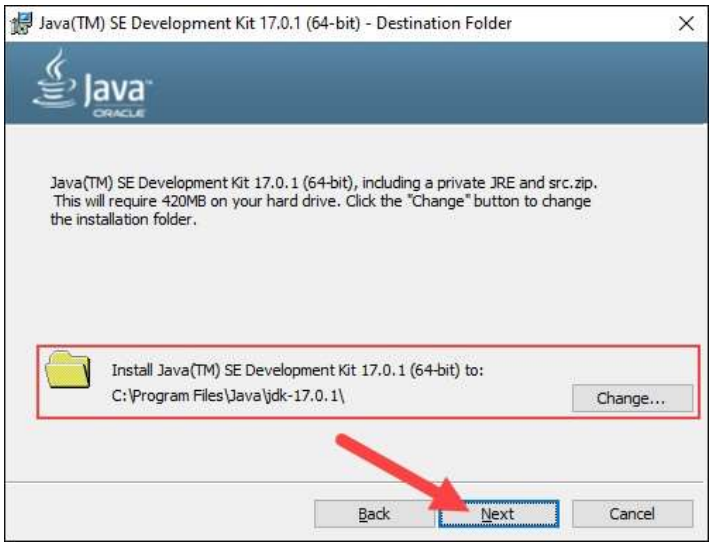
Follow the steps below:

Step 1: Run the Downloaded File
Double-click the **downloaded file** to start the installation.

Step 2: Configure the Installation Wizard
After running the installation file, the installation wizard welcome screen appears.

1. Click Next to proceed to the next step.

2. **Choose the destination folder for the Java installation files or stick to the default path. Click** Next **to proceed**



3. **Wait for the wizard to finish the installation process until the *Successfully Installed* message appears. Click** Close **to exit the wizard.**

Set UP Java Environment

Go to MyComputer properties -> advanced tab -> environment variables -> new tab of user variable -> write path in variable name -> write path of bin folder in variable value -> ok -> ok -> ok

1) Go to My Computer properties
2) Click on the advanced tab
3) Click on environment variables
4) Select the path in the System Variable and click on Edit
5) Copy the path of Java bin folder
6) Paste path of bin folder in the Environment Variable
7) Click on ok button 8) Click on ok button

2. Install java editor (Eclipse for Enterprise Java) and configure workspace

5 Steps to Install Eclipse

1. Download the Eclipse Installer

Download Eclipse Installer from http://www.eclipse.org/downloads

Eclipse is hosted on many mirrors around the world. Please select the one closest to you and start to download the Installer

2. Start the Eclipse Installer executable

For Windows users, after the Eclipse Installer executable has finished downloading it should be available in your download directory. Start the Eclipse Installer executable. You may get a security warning to run this file. If the Eclipse Foundation is the Publisher, you are good to select Run. For Mac and Linux users, you will still need to unzip the download to create the Installer. Start the Installer once it is available.

3. Select the package to install

The new Eclipse Installer shows the packages available to Eclipse users. You can search for the package you want to install or scroll through the list.

Select and click on the package you want to install.

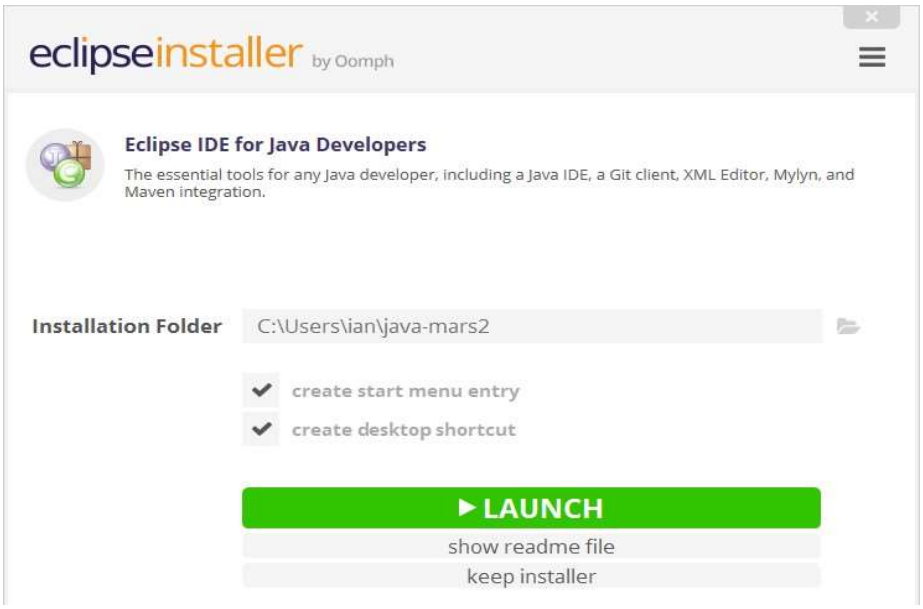
4. Select your installation folder

Specify the folder where you want Eclipse to be installed. The default folder will be in your User directory.

Select the 'Install' button to begin the installation.

5. Launch Eclipse

Once the installation is complete you can now launch Eclipse. The Eclipse Installer has done it's work. Happy coding.

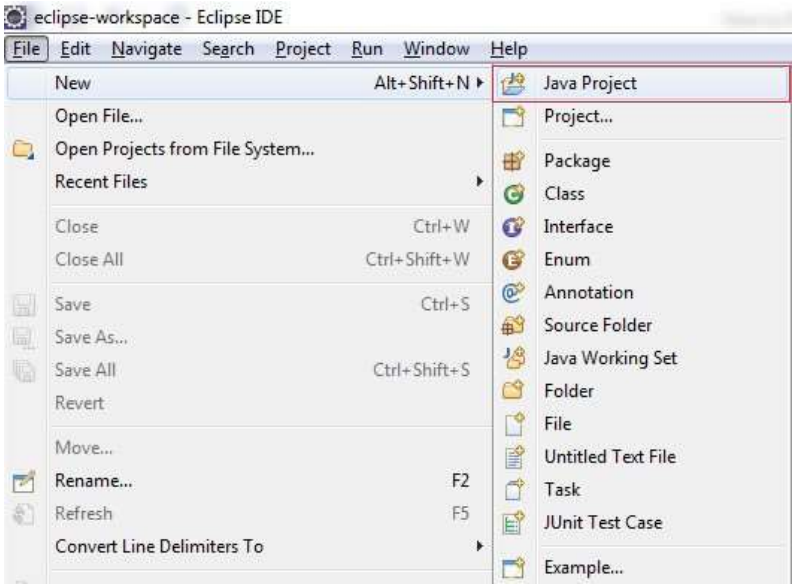


3. Execution of first java program

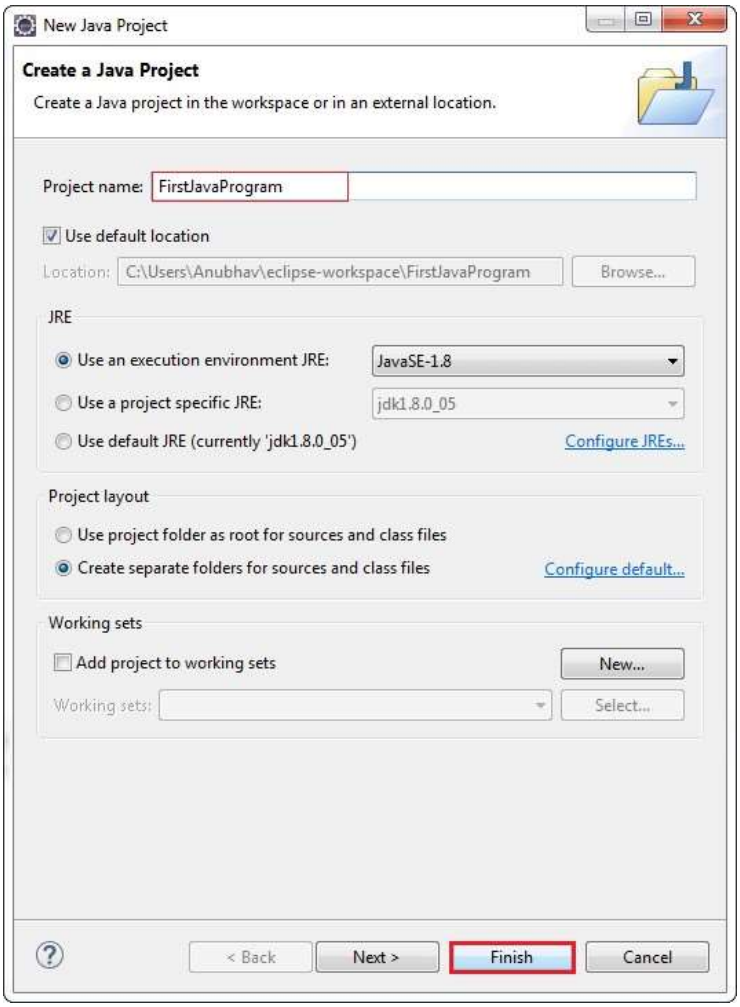
How to Run Java Program in eclipse

In this section, we learn how to run a Java program in eclipse step by step.

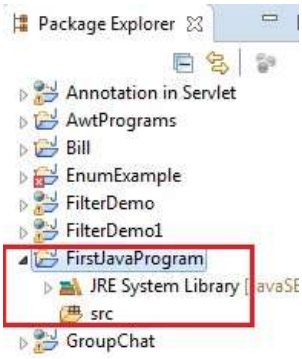
Step 1: Open Eclipse and click **File > New > Java Project**.



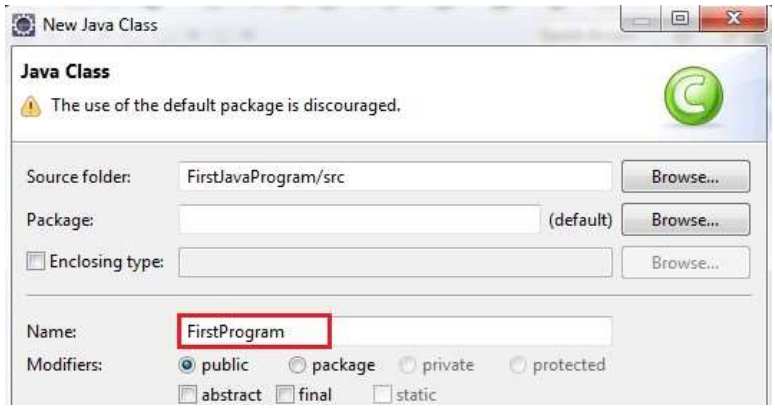
Step 2: Provide the **Project Name** and click on the **Finish** button.



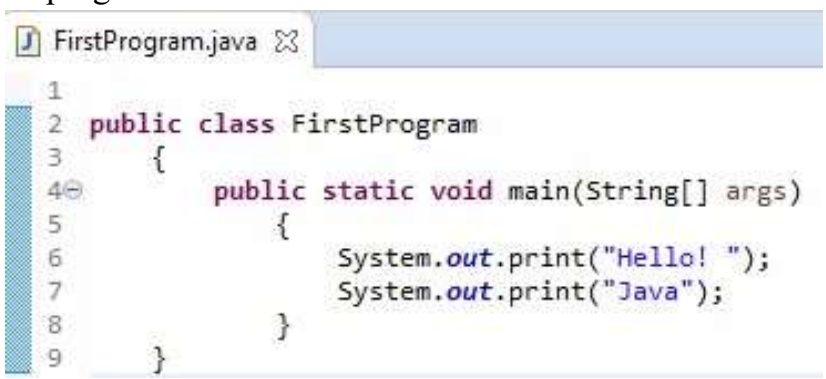
Step 3: In the **Package Explorer** (left-hand side of the window) select the project which you have created.



Step 4: Right-click on the **src** folder, select **New > Class** from the submenu. Provide the **Class name** and click on **Finish** button.

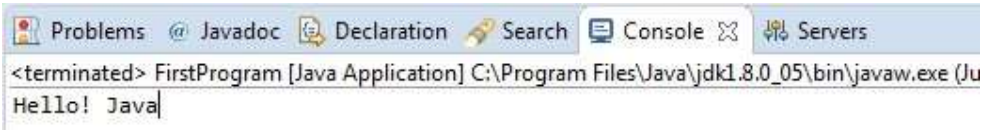


Step 5: Write the program and save it.



Step 6: Now, press **Ctrl+F11** or click on the **Run** menu and select **Run** or click on Run button.

Step 7: **Output**



4. Java code execution process

Consider the program '*Hello.java*' that we explained in the last chapter of this tutorial.

Source file: Hello.java

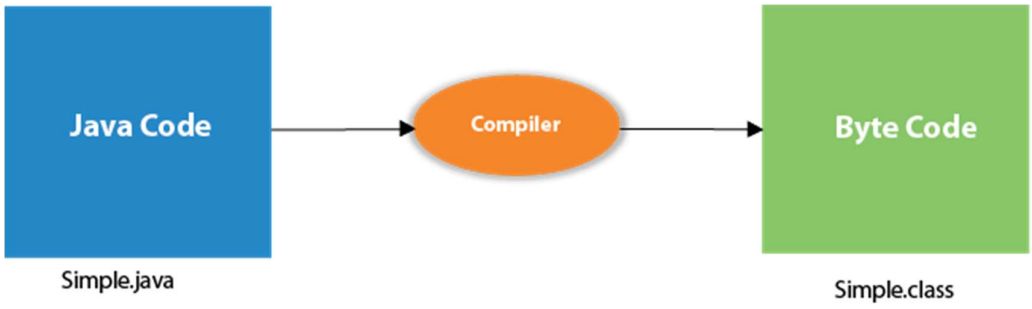
```
public
class Hello
{
        public static void main(String[] args)
        {
                System.out.print("My first program in java");
        }

}
```

Let's now understand what happens during compile and run time of the above code.

Internal working during compile time

- When you compile the **Hello.java** file then the compiler creates a **.class file**
- **.class file** is created with the name **Hello.class**
- **Hello.class** contains **byte codes**



Byte code (.class file) is specialized set of instruction that JVM can read.

Thus this .class file consisting of byte code can be transported to other system, irrespective of the operating system it has, and could be executed comfortably without any issue.

This is what makes java a **portable language**. Because of the same reason java is also popularly known as **Write once, run anywhere (WORA) language**.

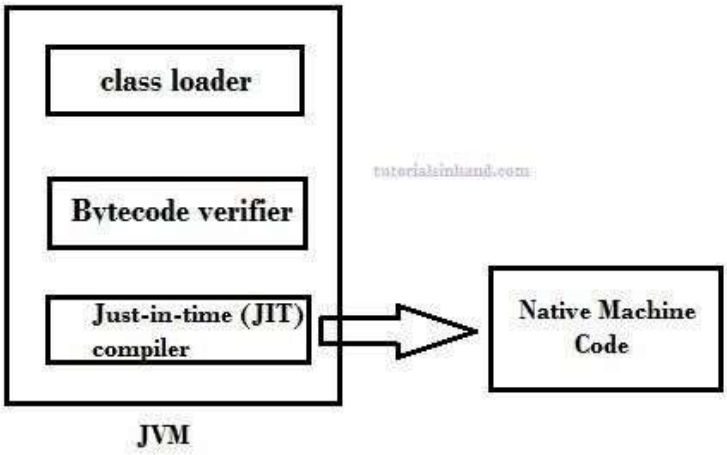
JVM is capable of reading, verifying the bytecode.

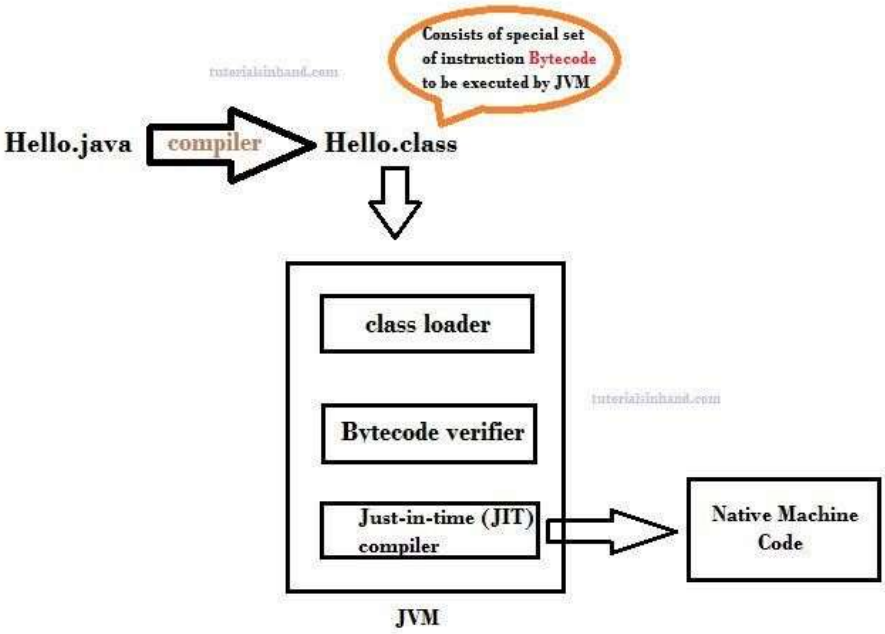
Any computer system on which JVM is installed will be capable of executing the compiled .class file (consisting of bytecode).

Internal working during run time

During run or execution time following tasks are performed within JVM:

- Class loader loads the .class files containing bytecode to the memory.
- Next the bytecode is verified and checked for any errors or bugs which may result in program exhibiting anamolous behaviour.
- At last the just-in-time compiler converts the bytecode into machine code.



Compile + Run diagram of java code

When we combine both compilation and execution of java program together, we get the complete internal process the java program goes through.